

28th Experiment

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.metrics import accuracy_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.optimizers import Adam

# Generate Multi-Class Dataset
X, y = make_blobs(
    n_samples=500,
    centers=3,
    n_features=2,
    cluster_std=1.5,
    random_state=42
)

# One-Hot Encoding
y_cat = to_categorical(y)

# Build Neural Network
model = Sequential([
    Dense(16, activation='relu', input_shape=(2,)),
    Dense(8, activation='relu'),
    Dense(3, activation='softmax')
])
```

```
# Adam Optimizer with Learning Rate 0.001
```

```
optimizer = Adam(learning_rate=0.001)
```

```
# Compile Model
```

```
model.compile(
```

```
    optimizer=optimizer,
```

```
    loss='categorical_crossentropy',
```

```
    metrics=['accuracy']
```

```
)
```

```
# Train Model
```

```
history = model.fit(
```

```
    X,
```

```
    y_cat,
```

```
    epochs=50,
```

```
    verbose=0
```

```
)
```

```
# Training Loss Graph
```

```
plt.figure(figsize=(5,3))
```

```
plt.plot(history.history['loss'])
```

```
plt.title("Training Loss")
```

```
plt.xlabel("Epoch")
```

```
plt.ylabel("Loss")
```

```
plt.grid(True)
```

```
plt.show()
```

```
# Prediction
```

```
pred = model.predict(X)
```

```
pred_class = np.argmax(pred, axis=1)
```

```
# Accuracy
```

```
acc = accuracy_score(y, pred_class)
```

```
print("Accuracy:", round(acc*100,2), "%")
```

```
# Dataset Visualization
```

```
plt.figure(figsize=(5,4))
```

```
plt.scatter(X[:,0], X[:,1], c=y)
```

```
plt.title("Multi-Class Data (ReLU, LR=0.001)")
```

```
plt.xlabel("Feature 1")
```

```
plt.ylabel("Feature 2")
```

```
plt.show()
```

Output:

